

120 Years of Olympic History: Participation and Medal Trends

Vibhor Dabas

Shravani Jadhav

Minh Chung

Table of contents

1	Executive Summary	1
2	Introduction	1
3	Research Questions	2
4	Methodology	2
4.1	Data source:	2
4.2	Visualisation:	3
4.3	Statstical Analysis:	3
5	Results	4
5.1	Visulization	4
5.2	Statistical Analysis	5
6	Conclusion and Discussion	12
7	Limitations	12
	References	13

1 Executive Summary

This report looks at 120 years of Olympic history – from the 1896 Athens Games to the 2016 Rio Games - to analyze participation and medal-winning trends. The analysis shows a major rise in the number of women athletes over time, and a continued dominance of wealthy countries in the medal tallies. Australia has a high Olympic performance relative to its population. The

findings highlight the evolving inclusivity of the Olympics and the impact of national resources on sporting success.

2 Introduction

The Olympic Games is the most popular international sports competition that gathers the best sportsmen from all the world every four years. Since their modern revival in Athens in 1896, the Games have expanded enormously in scale, diversity and global importance. Studying historical trends in Olympic participation and performance can shed light on wider trends in gender equality, national development and sporting investment. This report looks at athlete-level data over 120 years of Olympic history to bring out key trends in participation and medal outcomes.

The dataset, from Kaggle (Griffin 2018), contains over 271,000 observations of athletes, sports, events, and medal results from Summer and Winter Games. The analysis is restricted to the Summer Olympics to ensure comparability of similar events.

By analysing both visual and statistical trends, this report aims to answer key questions about how have participation and medal-winning patterns changed across Olympic Games from 1896 to 2016, with a focus on gender representation and top-performing nations

3 Research Questions

The analysis is guided by three key questions:

- How has overall athlete participation in the Summer Olympic Games changed over time?
- How has female representation among Olympic athletes changed over time?
- Has the proportion of Olympic medals won by female athletes increased over time?
- Has the distribution of Olympic medals among nations become more or less concentrated over time?

4 Methodology

4.1 Data source:

The dataset is sourced from Kaggle and acknowledged as The Olympic data on www.sports-reference.com, which contains information on athletes, their participation in Olympic Games, and the medals they won from 1896 to 2016. The dataset includes variables:

- ID: Unique number for each athlete
- Name: Athlete's name
- Sex: M or F
- Age: Integer
- Height: In centimeters
- Weight: In kilograms
- Team: Team name
- NOC: National Olympic Committee 3-letter code
- Games: Year and season
- Year: Integer
- City: Host city
- Sport: Sport
- Event: Event
- Medal: Gold, Silver, Bronze, or NA

4.2 Visualisation:

This data analysis makes use of the “120 Years of Olympic History” dataset found on Kaggle and has a total of 271,116 observations in 15 different variables that cover the period from the Athens Games in 1896 to the Rio Games in 2016 (Griffin 2018). Data is related to individual athletes such as their names, sex, age, country, sport, event, and the medals won. Only the Summer Olympic Games were considered in order to ensure consistency in the types of events being compared since there are differences between the Summer and Winter Olympic Games in terms of sports, countries that participate, and numbers of athletes. The Age, Height and Weight variables had missing values which were not necessary for the analysis. Medal values of NA indicate non-medallists and were excluded from medal counts.

4.3 Statistical Analysis:

Moreover, to analyse and infer how participation and medal-winning patterns have changed across the Olympic Games from 1896 to 2016, we examine four key aspects: overall participation, female representation, female medal success, and medal concentration among nations. Statistical models are used to assess whether these measures exhibit significant trends over time.

1. Overall participation patterns over time: To determine whether Olympic participation has increased over time, a linear regression model is fitted: $Participation_t \sim \beta_0 + \beta_1 * Year + \epsilon_t$ where $Participation_t$ is the number of athletes participating in each Year. A significant positive coefficient for Year (β_1) would indicate an increase in participation over time.
2. Female Representation over time: To assess whether female representation has changed over time, the proportion of female athletes is analysed using a binomial logistic regression model: $\log\left(\frac{p_t}{1-p_t}\right) \sim \beta_0 + \beta_1 * Year + \epsilon_t$ where p_t is the proportion of female athletes in each Year. The model is fitted using the yearly counts of female and male athletes. A significantly positive estimate of β_1 would suggest an increasing trend in female representation over time.
3. Female Medal Success over time: To investigate whether female athletes have become more successful over time, the proportion of medals won by women is analysed using a binomial logistic regression model: $\log\left(\frac{p_t}{1-p_t}\right) \sim \beta_0 + \beta_1 * Year + \epsilon_t$ where p_t is the proportion of medals won by female athletes in each Year. The model is fitted using the yearly counts of medals won by female and male athletes. A significantly positive estimate of β_1 would indicate an increasing trend in female medal success over time.
4. Medal Concentration Among Nations: To evaluate whether medal concentration among nations has changed over time, the Herfindahl-Hirschman Index (HHI) is calculated for each Year based on the distribution of medals among countries. A linear regression model is then fitted: $HHI_t \sim \beta_0 + \beta_1 * Year + \epsilon_t$ where HHI_t is the Herfindahl-Hirschman Index for each Year. A significant negative coefficient for Year (β_1) would suggest a decrease in medal concentration among nations over time, indicating a more even distribution of medals across countries.

5 Results

5.1 Visualization

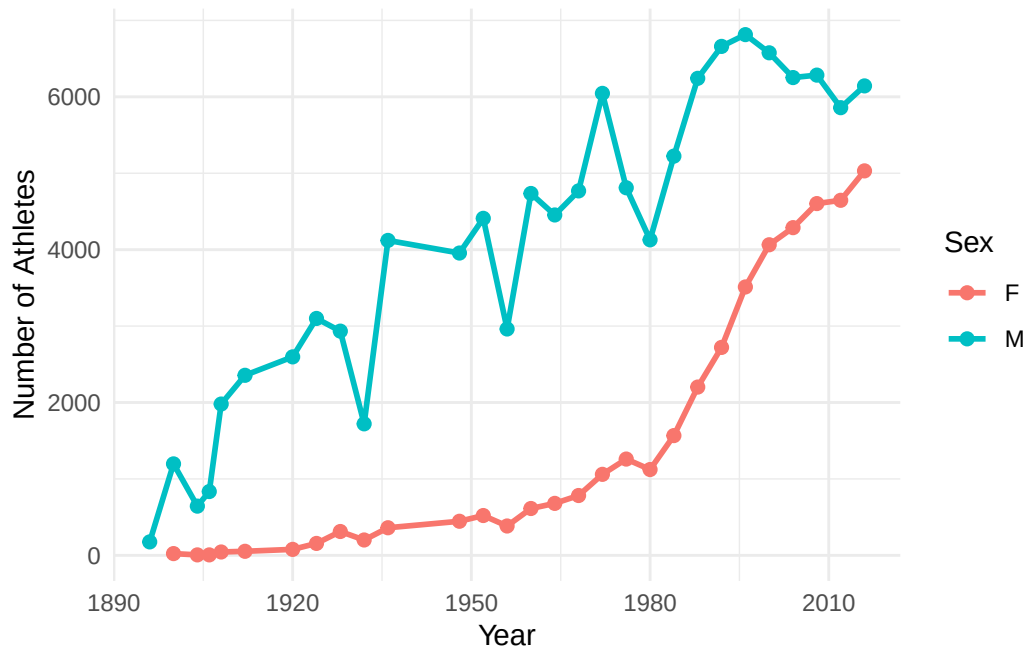


Figure 1: Athlete Participation by Gender in the Summer Olympics (1896–2016)

Table 1: Top 5 Countries by Total Medals (Summer Olympics)

Country	Total Medals
United States	4686
Soviet Union	2061
Germany	1687
Great Britain	1598
France	1408

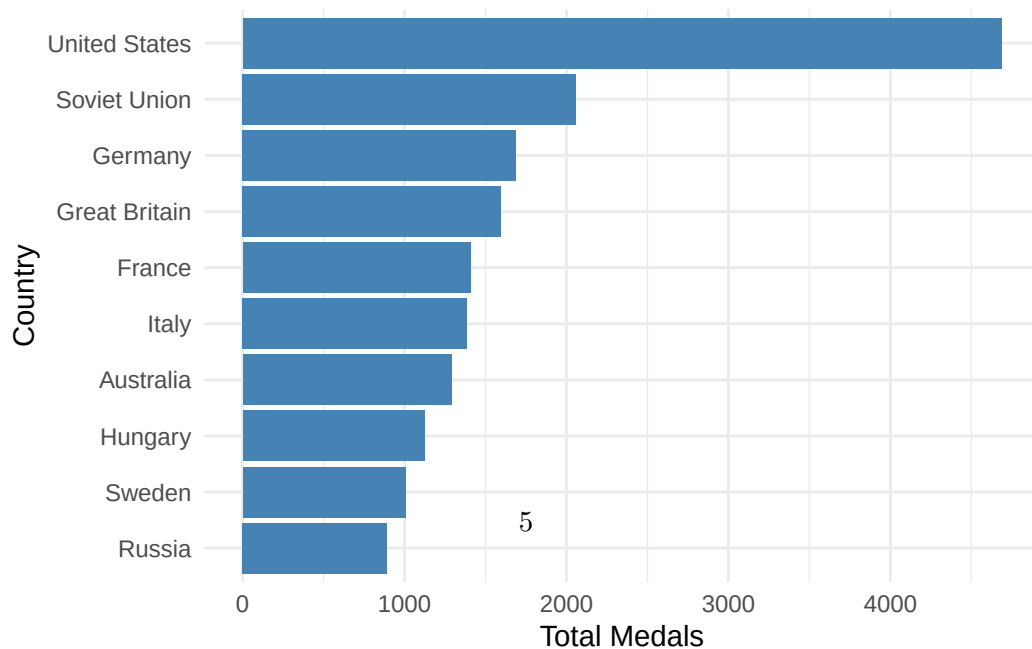
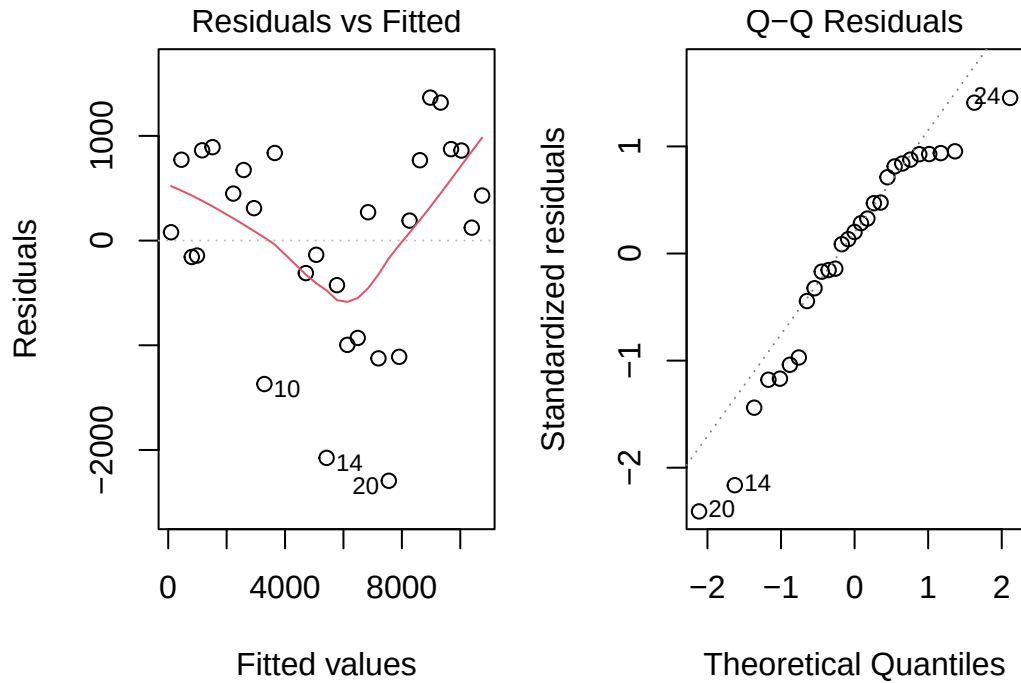


Figure 2: Top 10 Countries by Total Medal Count in the Summer Olympics (1896–2016)

Table 2

```
# A tibble: 1 x 8
```

	r.squared	adj.r.squared	sigma	statistic	p.value	df	df.residual	nobs
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<dbl>
1	0.925	0.922	976.	332.	1.07e-16	1	27	29



(a) Residuals vs Fitted

Figure 3: Residual plots for participation model

Table 3

```

Call:
glm(formula = cbind(Female, Male) ~ Year, family = binomial,
     data = gender_participation)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -6.500e+01  5.598e-01  -116.1   <2e-16 ***
Year          3.220e-02  2.811e-04   114.6   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 18562.75  on 28  degrees of freedom
Residual deviance:  471.84  on 27  degrees of freedom
AIC: 694.54

Number of Fisher Scoring iterations: 4

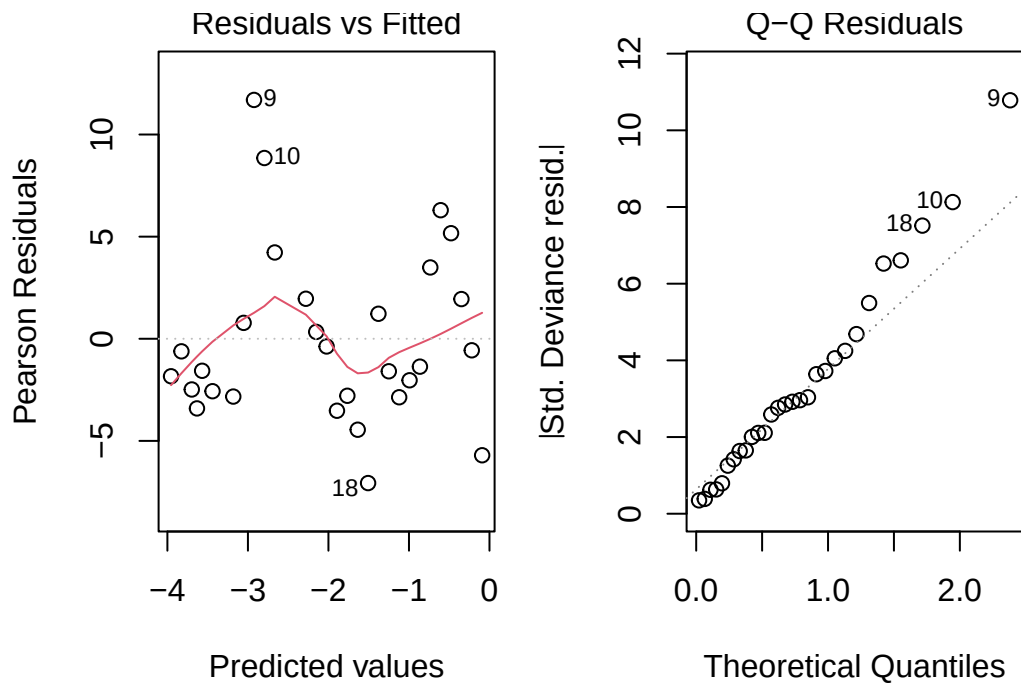
```

According to Table 2, the coefficient for Year is positive and statistically significant ($p < 0.001$), indicating that Olympic participation has increased over time. However, the relatively low R-squared value suggests that Year alone explains only a limited proportion of the variation in participation. Additionally, the residuals plots show clear patterns rather than random variation, indicating that the assumptions of linear regression may not be fully satisfied. Therefore, while there is evidence of a significant upward trend in participation, a more complex model may better capture the relationship over time.

2. Female Representation over time:

The binomial logistic regression indicates that female representation in the Olympic Games has increased significantly over time. Based on Table 3, the coefficient for Year is positive and highly significant ($p\text{-value} < 0.001$) suggesting that the proportion of female athletes has steadily increased throughout the study period. The large reduction in deviance from the null model (20058.38) to the fitted model (503.82) indicates that Year explains a substantial amount of the variation in female participation rates. The residual plots look reasonably random, indicating a good fit for the model. Therefore, there is strong evidence that female representation in the Olympics has increased over time.

3.



(a) Residuals vs Fitted

Figure 4: Residual plots for female representation model

Table 4

```
Call:
glm(formula = cbind(FemaleMedals, MaleMedals) ~ Year, family = binomial,
    data = gender_medals)
```

Coefficients:

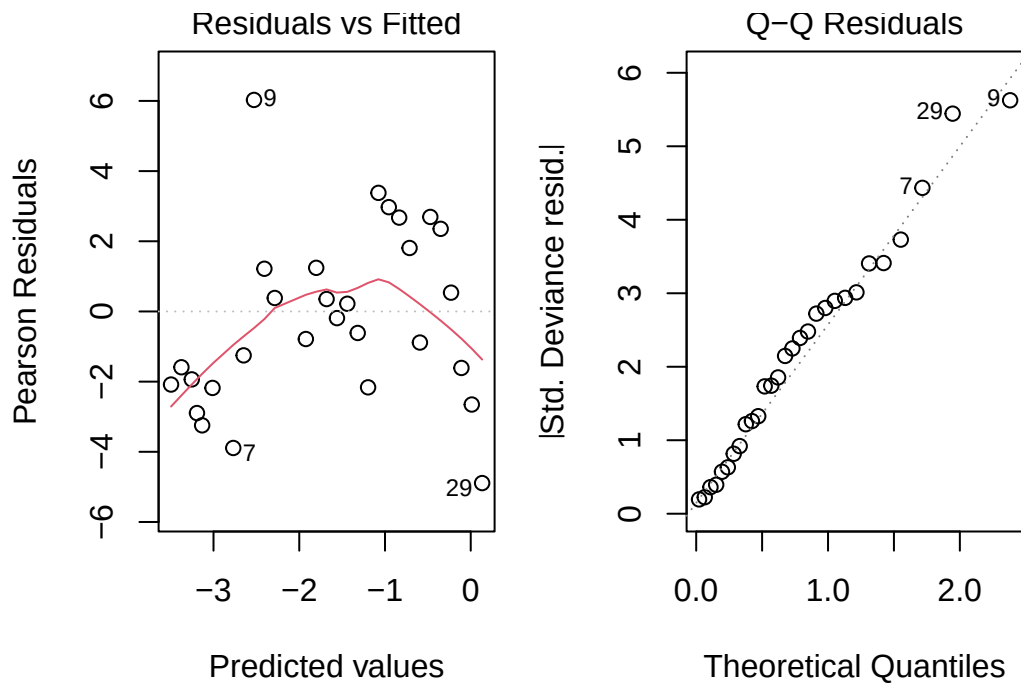
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-6.083e+01	1.011e+00	-60.16	<2e-16 ***
Year	3.024e-02	5.088e-04	59.43	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 5109.44 on 28 degrees of freedom
 Residual deviance: 181.18 on 27 degrees of freedom
 AIC: 370.36

Number of Fisher Scoring iterations: 4



(a) Residuals vs Fitted

Figure 5: Residual plots for female medal success model

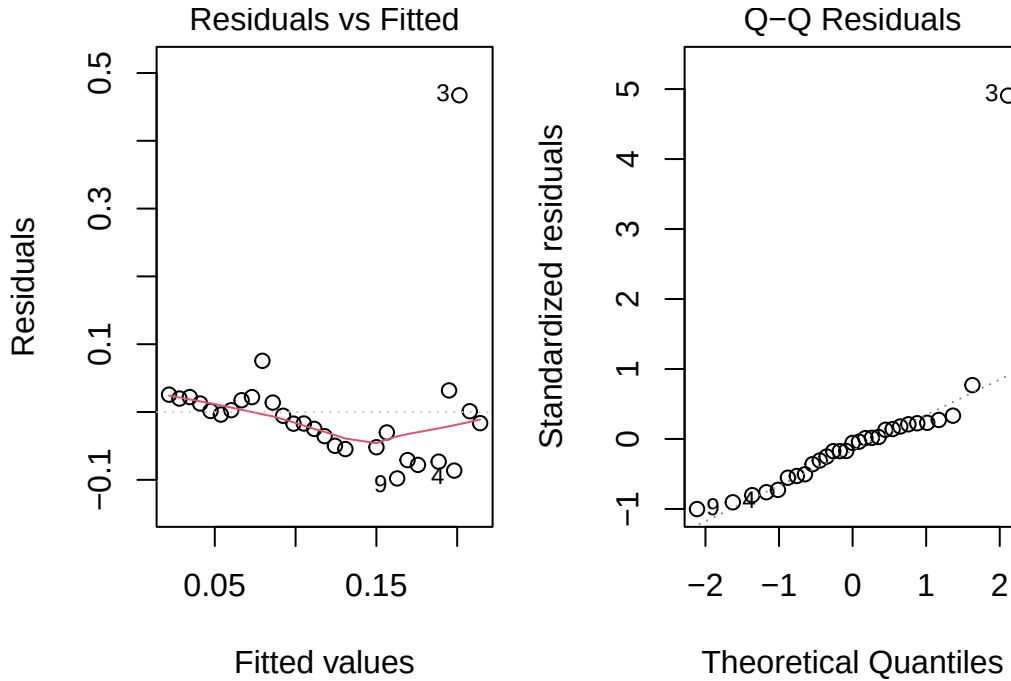
Table 5

```
# A tibble: 1 x 8
```

	r.squared	adj.r.squared	sigma	statistic	p.value	df	df.residual	nobs
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<dbl>
1	0.275	0.248	0.101	10.2	0.00352	1	27	29

The logistic regression shows a significant increase in female medal success over time (p-value < 0.001). Table 4 shows the positive coefficient indicating that the proportion of medals won by female athletes has steadily increased throughout Olympic history. Furthermore, the substantial reduction in deviance from the null model (5548.17) to the fitted model (187.68) indicates that Year explains a considerable amount of the variation in female medal success rates. These results provide strong evidence that women's contribution to overall Olympic medal success has increased over time.

4. Medal Concentration Among Nations:



(a) Residuals vs Fitted

Figure 6: Residual plots for HHI model

Table 5 indicates a significant decline in medal concentration over time. The coefficient for Year is negative and statistically significant with $\beta_1 = -0.00140$ and $p = 0.0017$, suggesting

that Olympic medals have become less concentrated among a small number of countries and more evenly distributed across participating nations. The model explains approximately 26.2% of the variation in the HHI ($R^2 = 0.262$), indicating that Year is a meaningful predictor of medal concentration, although other factors may also influence the distribution of medals. The residual plots satisfy the assumptions of linear regression, showing no clear patterns and suggesting a good fit for the model. Overall, these results provide strong evidence that medal concentration among nations has decreased over time, indicating a more competitive and inclusive Olympic Games.

6 Conclusion and Discussion

The analysis of Summer Olympic Games data from 1896 to 2016 reveals substantial changes in participation and medal outcomes over time. Athlete participation increased significantly throughout the study period, reflecting the expansion of the Olympic movement and the growing global interest in international sport. Female representation also rose considerably, with women accounting for an increasingly larger share of participants and medal winners. These findings indicate that the Olympic Games have become progressively more inclusive and have provided greater opportunities for female athletes to compete and succeed at the highest level.

The results also suggest that Olympic success has become less concentrated among a small group of nations. The decline in the Herfindahl–Hirschman Index over time indicates a more even distribution of medals across participating countries, pointing to increased competitiveness within the Games. While historically successful nations continue to achieve strong results, the broader distribution of medals suggests that more countries have been able to develop competitive sporting programs. Overall, the findings highlight the growth, inclusiveness, and increasing international competitiveness of the Summer Olympic Games over the past 120 years.

7 Limitations

This study has several limitations. The analysis focuses on long-term trends using Year as the primary explanatory variable and does not account for other factors that may influence participation and medal outcomes, such as population size, economic development, government investment in sport, host-country effects, or geopolitical events. In addition, the statistical models assume relatively simple relationships between Year and the outcomes of interest, which may not fully capture periods of rapid change in Olympic participation and performance. The measure of medal concentration also describes how medals are distributed among nations but does not identify the underlying causes of those patterns. Future research could incorporate

additional economic, demographic, and policy-related variables to provide a more comprehensive understanding of the factors shaping Olympic success.

References

Griffin, Randi. 2018. *120 Years of Olympic History: Athletes and Results*. <https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results>.